

UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF PHYSICS

CONTINUOUS ASSESSMENT TEST-CAT

MODULE: PHY1163-Physics for Engineers I /40marks

TARGET GROUPS: EPE, Level 1

Date: 13th July, 2022

Duration: 2hours

QUESTION N^o 1 /15marks

- a) In the freely falling motion a ball is thrown vertically upward. What are its velocity and acceleration when it reaches its maximum altitude? What is its acceleration just before it strikes the ground? /3marks
- b) A stone is thrown from the top of a building with an initial velocity of 20m.s^{-1} straight upward. The building is 50 m high, and the stone just misses the edge of the roof on its way down. Determine: /4marks
- i) The time needed for the stone to reach its maximum height,
 - ii) The maximum height,
 - iii) The velocity and position of the stone at $t = 5\text{s}$
- c) A stone is catapulted at time $t = 0\text{s}$, with an initial velocity of magnitude 20.0 m/s and at an angle of 40.0° above the horizontal.
- i) What are the horizontal x - and vertical y - components of its displacement from the catapult site (x_0, y_0) at time $t = 1.10\text{s}$? /4marks
 - ii) How far will the stone travel in the horizontal direction at that time? /2marks
 - iii) How long will it take the stone to turn back to the level of catapult site? /2marks

20-39

QUESTION N^o 2 /10marks

- A) Starting with the formula for the moment of inertia of a long thin rod rotated around an axis passing at one end perpendicular to its length $I = \frac{1}{3}ML^2$, prove that the moment of inertia of the same long thin rod rotated about an axis passing through its center perpendicular to its length is $I = \frac{1}{12}ML^2$. Where M is the mass of the rod, and L is its the length. /5marks
- B) A spherical ball of mass $M = 7.2\text{kg}$ and radius $R = 11\text{cm}$, starts from rest state and rolls without friction on a inclined plane by $\theta = 34^\circ$ to the horizontal and of length $l = 2.1\text{m}$. Determine:

$$v^2 = u^2 + 2as$$
$$y = y_0 + v_0 t + \frac{1}{2}gt^2$$
$$y = y_0 + \frac{(v_f + v_i)t}{2}$$
$$h = v_0$$

- a) The linear velocity V_C of the ball when it reaches the bottom of the slope. (Hint: use concept of the conservation of total mechanical energy, and noting that $I_C = \frac{2}{5}MR^2$) /3marks
- b) The total kinetic energy K of the ball at that position. /2marks

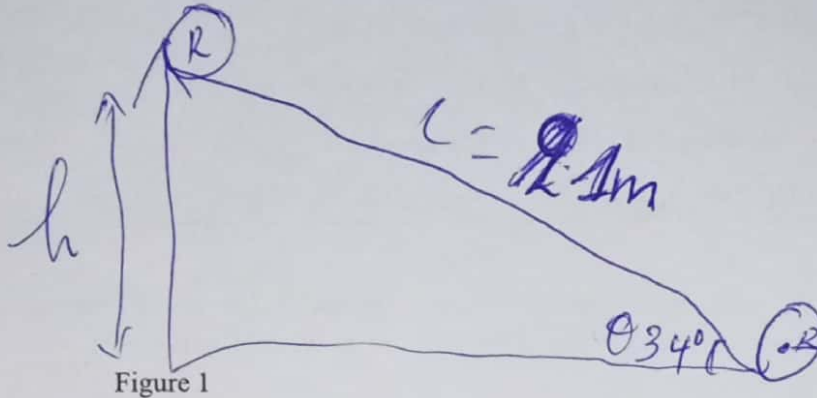


Figure 1

QUESTION N° 3 / 15marks

- A) State the work-energy theorem and give its mathematical expression. /2 marks
- B) Using the relationship between the conservative force and the potential energy function, find the conservative forces for the following potential energy functions: /3marks
- $U(x) = \frac{1}{2}kx^2$
 - $U(y) = mgy$
- C) A man pulls a wooden box on a frictionless floor with a force F of magnitude 50 N at an angle of 30° with the horizontal as shown in figure below.
- Draw a labeled free body diagram showing all the forces that act on the wooden box. /2marks
 - What is the magnitude of the work done by the weight of the wooden box? Explain briefly /3marks
 - Calculate the work done by the man on the wooden box as it is displaced 3 m to the right /5marks

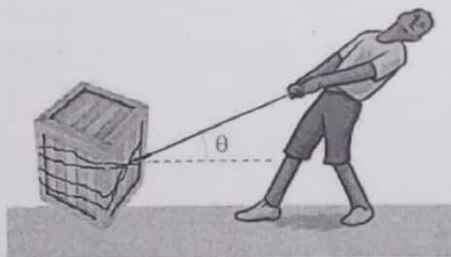


Figure 2

END OF QUESTIONS

$$K.E = \frac{1}{2}MV^2 + \frac{1}{2}I\omega^2$$

$$V_C = \frac{S}{t}$$

$$a_C = \frac{v^2}{r}$$

$$I_C \rightarrow V_C$$

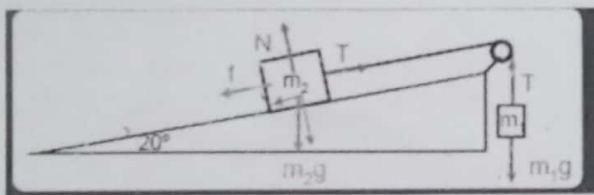
Question 1: a) What is the difference between fundamental and derived quantities?

b) Give at least three examples of each quantity cited at a). **(4marks)**

Question 2: A boat moves 2.0 km east then 4.0 km north, then 3.0 km west, and finally 2.0 km south. a) Draw each vector to scale with the tip of 1st to the tail of 2nd, the tip of 2nd to tail 3rd, and so on for others. **(4marks)**

b) Draw a resultant displacement from origin to tip of the last vector.
c) Finally, determine the resultant displacement analytically and its direction. **(4marks)**

Question 4: Two bodies, $m_1 = 1\text{ kg}$ and $m_2 = 2\text{ kg}$, are connected over a massless pulley. The coefficient of kinetic friction between m_2 and the incline is 0.1. The angle θ of the incline is 20° . Calculate: (a) Acceleration of the blocks. (b) The tension of the cord. **(4marks)**



Question 5:

- a) Define energy and Power
- b) Give at least three types of energy
- c) The driver of a 1 000 kg car traveling at a speed of $16,7 \text{ m}\cdot\text{s}^{-1}$ applies the car's brakes when he sees a red light. The car's brakes provide a frictional force of 8 000 N. Determine the stopping distance of the car. **(4marks)**

EPE2165—Exam #2

August 3, 2022

1. (20 points) The NMOS transistor in Figure 1 has a threshold voltage $V_t = 0.7V$, $\mu_n C_{ox} = 100 \mu A/V^2$, $L = 1 \mu m$, and $W = 32 \mu m$. If the channel-length modulation effect is neglected, calculate the value of R_D and R_S so that the transistor operates with a current $I_D = 0.4 mA$ at a voltage $V_D = 0.5V$.

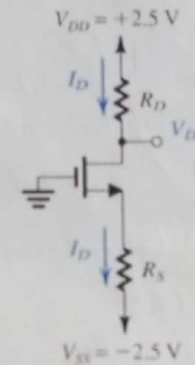


FIGURE 1

2. The BJT in Figure 2 has $\beta = 100$, $V_{CESat} = 0.3V$ and $V_{B(on)} = 0.7V$. Find V_E , V_C and I_B , I_C and the transistor's mode of operation:
- (a) (10 points) When $V_B = 0V$
- (b) (10 points) When $V_B = 3V$
- (c) (10 points) When $V_B = 5V$

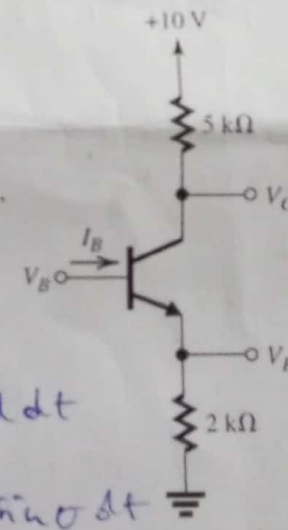


FIGURE 2

Handwritten calculations for problem 2:

$$I_{DC} = I_{AV}$$

$$I_{DC} = \frac{1}{T} \int_0^T i_D dt$$

$$X_{rms} = \sqrt{\frac{1}{T} \int_0^T x^2 dt}$$

$$I_{DC} = \frac{1}{2\pi} \int_0^{2\pi} i_D dt$$

$$I_{DC} = \frac{1}{2\pi} \int_0^{2\pi} I_m \sin \omega t dt$$

Other handwritten notes:

$$\omega_0 = 37 \text{ m/s}$$

$$\alpha = 53.1^\circ$$

$$t = 2 \times 10^{-6} \text{ s}$$



Module Title: PHYSICS FOR ENGINEERS I

Date: June 25, 2021

Module Code: PHY1163

Academic Year: 2020-2021 Class: Year 1 (GEO, SGE, ETE)

CONTINUOUS ASSESSMENT TEST (CAT) /30 marks

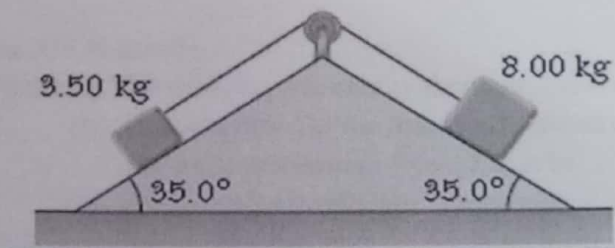
Duration: 2 hours

Question N°1 /6 marks

- (a) Write down the four kinematic equations that are most used to solve problems involving one-dimensional motion at constant acceleration and explain parameters involved. **2 marks**
- (b) A plane touches the ground for landing with a speed of 100 m/s and slow down at a rate of 5 m/s^2 before it stops.
- (i) What is the time it takes the plane to stop? **2 marks**
- (ii) Can this plane safely land on a small airport when the runway is only 0.8 km long? **2 marks**

Question N°2 /7 marks

- (a) State the Newton's laws of motion (First, Second and third Newton's law) **2 marks**
- (b) Two blocks of mass $m_1=3.5 \text{ kg}$ and $m_2=8.0 \text{ kg}$ are connected by a massless string that passes over a frictionless pulley (see Figure below). Given that the inclines are frictionless, $\theta=35^\circ$ and $g=9.8 \text{ m/s}^2$;



Find:

- (i) The magnitude of the acceleration vector of each block ($a_1=a_2$). **2 marks**

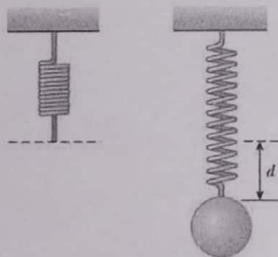
- (ii) The magnitude of the common tension vector force in the string ($T_1 = T_2$). **1 mark**
- (iii) The common value of the speed of each block at time $t = 2\text{s}$. **1 mark**
- (iv) The magnitude of the normal reaction forces exerted on the first block and on the second block ($N_1 \neq N_2$). **1 mark**

Question N°3 /5 marks

- (a) State Newton's second law for an object moving on a circle. **/2marks**
- (b) A civil engineer wishes to design a curved exit ramp for a highway in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a ramp is usually *banked*; this means the roadway is tilted toward the inside of the curve. Suppose the designated speed for the ramp is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 50.0 m . At what angle should the curve be banked? **3 marks**

Question N°4 /4 marks

- (a) State Hooke's law and explain why the spring force is sometimes called a restoring force. **2 marks**
- (b) A common technique used to measure the force constant of a spring is demonstrated by the setup in Figure below. The spring is hung vertically, and an object of mass m is attached to its lower end. Under the action of the "load" mg , the spring stretches a distance d from its equilibrium position.



- (i) If a spring is stretched 2.0 cm by a suspended object having a mass of 0.55 kg , what is the force constant of the spring? **1 mark**
- (ii) How much work is done by the spring as it stretches through this distance? **1 mark**

179.56
490

Question N°5 /8 marks

- (a) State: (i) The work-kinetic energy theorem. **1 mark**
- (ii) The condition for the force to be conservative and show that friction force is a non-conservative force. **2 marks**
- (iii) The law of conservation of mechanical energy. **1 mark**
- (b) A 15 kg block is launched upward on an inclined plane forming an angle of 37° with the horizontal. The block moves a distance of 3.0 m on the plane before stopping. The coefficient of friction between the block and the plane is 0.25 . Calculate: (i) The increase of potential energy of the block; (ii) The work done by the force of friction; (iii) The starting (initial) speed of the block; (iv) The speed of the block when it comes back to its point of departure while descending. **4 marks**

$$T \sim \sqrt{\frac{k}{m}}$$

- ii) State the Newton's laws of motion **1.5 marks**
- iii) What are conditions of static equilibrium for a rigid body **1 marks**
- iv) Differentiate between a laminar flow and a turbulent flow **1 marks**
- b) i) A block of mass $m = 1.5 \text{ kg}$ hangs at the end of a (weightless) cord suspended from the ceiling. What is the tension in the cord, and with what force does the cord pull down on the ceiling. **2 marks**
- ii) A metal rod that is 4.00 m long and 0.50 cm^2 in cross-sectional area is found to stretch 0.20 cm under a tension of 5000 N . What is Young's modulus for this metal? **1.5 marks**
- c) When a body of unknown mass is attached to an ideal spring with force constant 120 N/m , it is found to vibrate with a frequency of 6.00 Hz . Find (i) the period of the motion; (ii) the angular frequency; (iii) the mass of the body. **1.5 marks**

Question N⁰2/10 marks

- a) An automobile initially moving at 30 m/s accelerates uniformly at 5 m/s^2 . How fast is it moving after 3 s ? **1 marks**
- b) A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0 \text{ m/s}$ at an angle $\alpha = 53.1^\circ$. **3 marks**
- (i) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00 \text{ s}$.
- (ii) Find the time when the ball reaches the highest point of its flight, and its height at this time.
- (iii) Find the horizontal range.

- c) At a certain point in a horizontal pipeline the water's speed is 2.50 m/s and the gauge pressure is $1.80 \times 10^4 \text{ Pa}$. Find the gauge pressure at a second point in the line if the cross-sectional area at the second point is twice that at the first. **1 marks**

SECTION B: FROM THIS SECTION, CHOOSE ONLY TWO QUESTIONS

marks

/ 30

Question N⁰3 / 15 marks



UNIVERSITY OF
RWANDA

DEPARTMENT OF PHYSICS

FIRST TRIMESTER EXAMINATIONS – ACADEMIC YEAR 2019-2020

DEPARTMENT: PHYSICS

LEVEL OF STUDY: Year 1

STUDY GROUP: CE_CE, CE_EE, CE_GE, CE_SE, CE_WE, SGE, 1

CS_Huye, MEE_PROD_Gako, IS_Huye, IT_Huye, CS_Musanze, EEE_EPE,
EEE_ETE

MODULE CODE: PHY1163

MODULE NAME: Physics for Engineers I

DATE: 17/02/2020

DURATION: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains 5 questions to be answered to in 2 Sections: Section A and Section B.
2. Section A is compulsory; you must answer to all the questions.
3. Section B allows a choice; select any 2 questions from the 3 questions given and answer to them.
4. Write both your Registration and your Examination Numbers on the Examination Booklets provided.
5. Write all your answers only in the examination booklet provided.
6. Do not write on this question paper.
7. Start each question on a fresh page.
8. No unauthorized material is allowed in the examination room.

SECTION A: COMPULSORY QUESTIONS / 20 marks

Question N^o1/10 marks

- a) i) Using the method of dimensional analysis, show that the period of oscillation of a mass suspended on spiral spring is proportional to the spring constant and the mass according to the following equation 1.5 marks

SECTION A: COMPULSORY QUESTIONS [20 marks]

QUESTION 1 [12]

- a) State:
- Newton's laws of motion;
 - the work-kinetic energy theorem. (4 marks)
- b) Two blocks of mass $m_A = 3.5 \text{ kg}$ and $m_B = 8 \text{ kg}$ are connected by a massless string that passes over a frictionless pulley (Fig.1). Given that the inclines are frictionless and $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$; Find
- The magnitude of the normal reaction forces exerted on the block A and on the block B.
 - The magnitude of the acceleration vector of each block;
 - The magnitude of the common tension vector force in the string;
 - The common value of the speed of each block at time $t = 2 \text{ s}$. (8 marks)

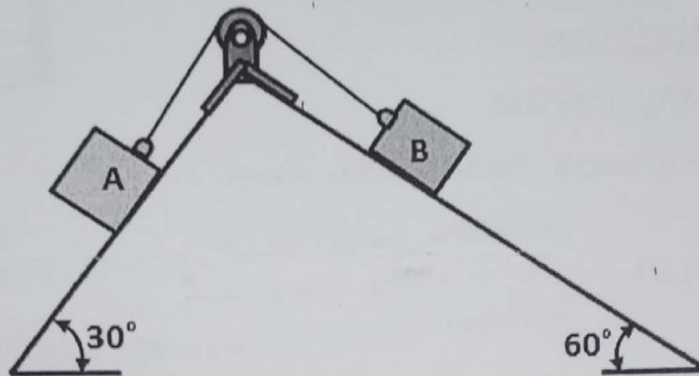


Figure 1: Two attached blocks sliding on inclined planes

QUESTION 2 [8]

A 15 kg block is launched upward on an inclined plane forming an angle of 37° with the horizontal. The block moves a distance of 3 m on the plane before stopping. The kinetic coefficient of friction between the block and the plane is 0.25 . Calculate:

- The increase of potential energy of the block;
- The work done by the force of friction;
- The starting (initial) speed of the block;
- The speed of the block when it comes back to its point of departure while descending.

SECTION B: YOU CHOOSE
marks]
QUESTION 3 [15]
a) A block

SECTION B: YOU CAN ATTEMPT ANY TWO QUESTIONS [30 marks]

QUESTION 3 [15]

- a) A thin rod of length $L = 60 \text{ cm}$ has a mass $M = 300 \text{ g}$;
- What is its linear mass density ρ (in SI units)?
 - Find its moment of inertia about an axis through its center of mass;
 - Find its moment of inertia about an axis through one of its ends. **(6 marks)**
- b) Consider a rigid body that can translate and rotate about an arbitrary axis under the action of two forces F_1 and F_2 applied on it. What are the conditions for this body to be in static equilibrium? (ii) The system in Fig.2 is in static equilibrium. The angles $\theta_1 = 60^\circ$ and $\theta_2 = 20^\circ$ and the ball has mass $M = 2 \text{ kg}$. Find the tensions T_1 and T_2 in the strings. **(9 marks)**

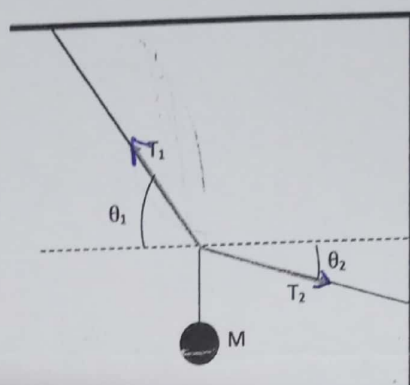


Figure 2

QUESTION 4 [15]

- a) On a very slippery road, a car A of mass $m_A = 1000 \text{ Kg}$ with initial speed of $v_A = 24 \text{ m/s}$ collides head-on with car B initially at rest. If the cars lock together and travel at 10 m/s after collision, find:
- The mass of car B?
 - The loss in total kinetic energy of the system during the collision. **(5 marks)**
- b) A ball with mass 300 g moving on a frictionless linear track at an initial speed of 1.2 m/s undergoes an elastic collision with a ball at rest of unknown mass. After the collision, the first ball continues in its original direction at 0.6 m/s .
- What is the mass of the second ball?
 - What is its speed after the impact? **(8 marks)**
- c) There is a direct relationship between the net force, the change in linear momentum and the time interval the action takes, as follows: $F_{\text{net}} = \Delta P / \Delta t$. To take a penalty shot, a professional football player kicks a ball of mass 400 g from rest with a net force of 100 N in a time interval of 0.1 s . At what velocity will it be launched? **(2 marks)**

QUESTION 5 [15]

- a) An office window has dimensions 3.0 m by 2.0 m. As a result of the passage of a storm, the outside pressure drops to 0.96 atm, but inside the pressure is held at 1 atm. What net force pushes out on the window? $1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$. (2 marks)
- b) The maximum depth of Kivu lake is about 475 m. Assuming that the Kivu Lake water has a uniform density of $1,01 \text{ kg/m}^3$, find the hydrostatic pressure (in atmospheres) that an object at its bottom has to withstand. (3 marks)
- c) i) Explain the next terms related to the elastic properties of rigid bodies: stress, strain, Young's modulus, yield strength (elastic limit).
- ii) Figure 3 shows the stress-strain curve for quartzite. What are the Young's modulus and the approximate yield strength for this material? (10 marks)

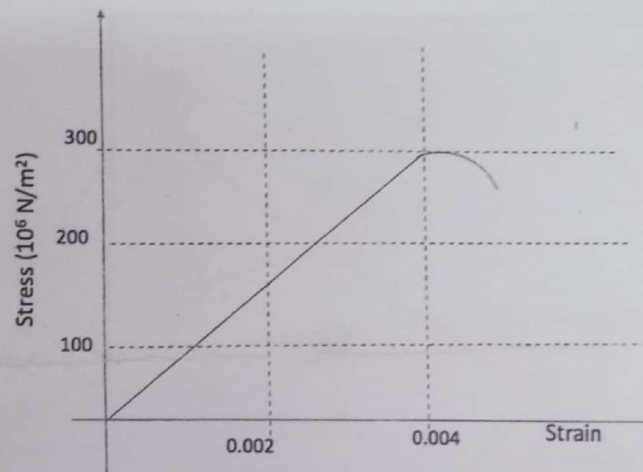


Figure 3

END OF QUESTIONS

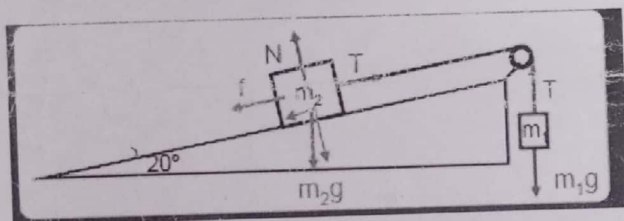
Question 1: a) What is the difference between fundamental and derived quantities?

b) Give at least three examples of each quantity cited at a). (4marks)

Question 2: A boat moves 2.0 km east then 4.0 km north, then 3.0 km west, and finally 2.0 km south. a) Draw each vector to scale with the tip of 1st to the tail of 2nd, the tip of 2nd to tail 3rd, and so on for others. (4marks)

b) Draw a resultant displacement from origin to tip of the last vector.
c) Finally, determine the resultant displacement analytically and its direction. (4marks)

Question 3: Two bodies, $m_1 = 1\text{ kg}$ and $m_2 = 2\text{ kg}$, are connected over a massless pulley. The coefficient of kinetic friction between m_2 and the incline is 0.1. The angle θ of the incline is 20° . Calculate: (a) Acceleration of the blocks. (b) The tension of the cord. (4marks)



Question 4:

a) Define energy and Power

b) Give at least three types of energy

c) The driver of a 1 000 kg car traveling at a speed of $16,7\text{ m}\cdot\text{s}^{-1}$ applies the car's brakes when he sees a red light. The car's brakes provide a frictional force of 8 000 N. Determine the stopping distance of the car. (4marks)